

TrES-1b

R-1537

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_140608 · created 2026-08-01T18:48:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456816.8326 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.014
Transit depth (Rp/Rs)^2	2.09 +/- 0.4 [%]
Orbital Inclination (inc)	89.74 +/- 0.97
Airmass coefficient 1 (a1)	1.0114 +/- 0.0095
Airmass coefficient 2 (a2)	-0.012 +/- 0.052
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#1 - [493, 167]
Optimal Aperture	3.6
Optimal Annulus	10.08
Transit Duration (day)	0.0872 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.145 $\pm$ 0.014 vs 0.1358 (deviation 0.66 σ)
Duration fit vs published	0.0872 d vs 0.1042 d (deviation 9.43 σ)
Mid-time O–C	2.2 min
Depth SNR	10.4
Residual scatter	0.94 %

Light curve

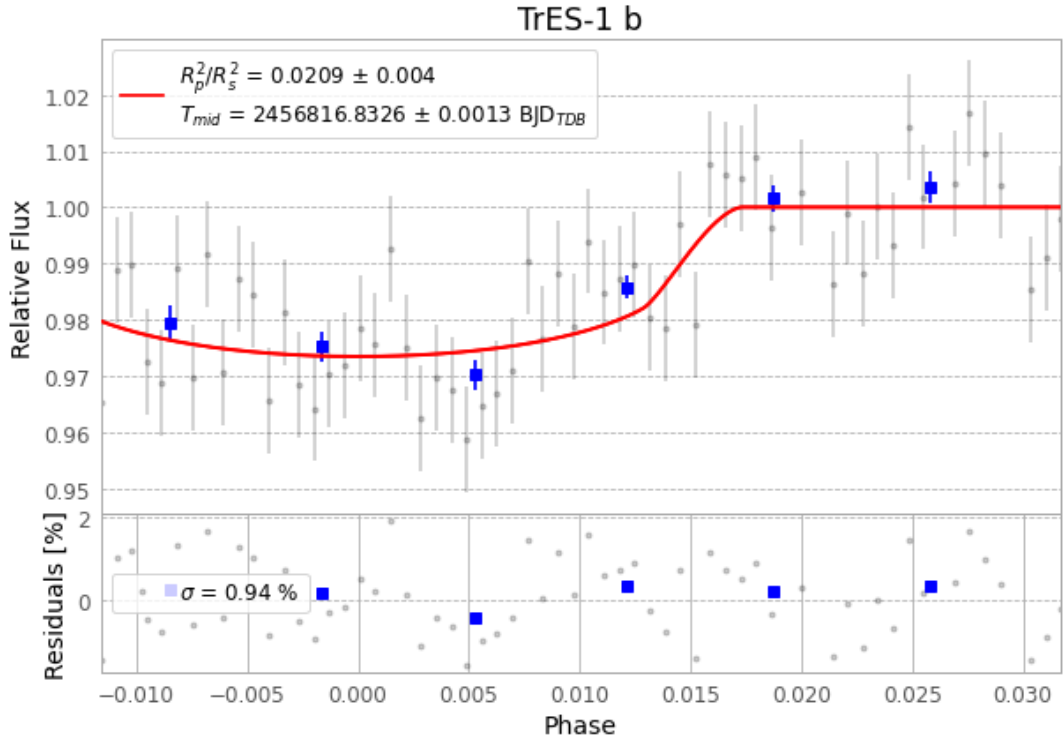


Figure 1 — FinalLightCurve_TrES-1 b_2014-06-08.png (SHA-256 81fc21e660c713ed...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f27233e0c8aaf50...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), TRES-1140608070312.FITS → TRES-1140608101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-140608.log (8712ac4d1512...), FinalLightCurve_TrES-1 b_2014-06-08.png (81fc21e660c7...), FinalLightCurve_TrES-1 b_2014-06-08.csv (a5472d7cd8e6...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 18:48Z.